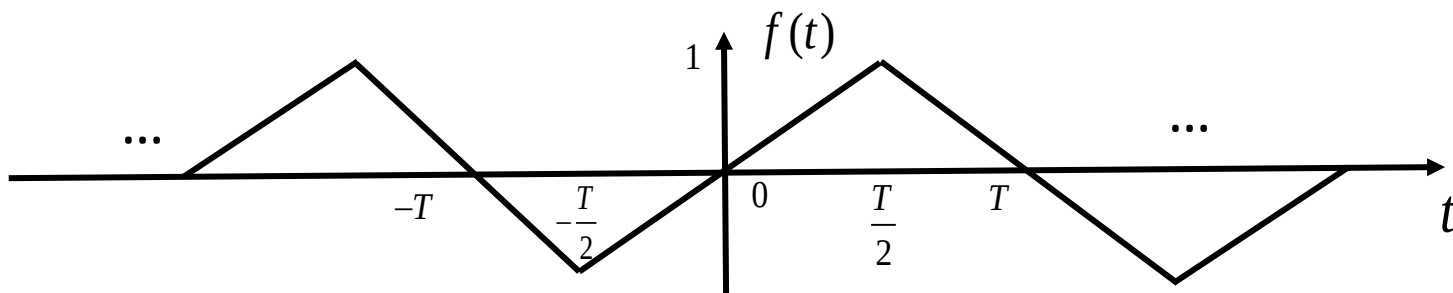


Problem Class 2

(Chapters 3 & 4)

Problem 3-1: The zero components of the Fourier series in trigonometric form for the periodic signal shown in the figure are ()



A

$$a_n = 0 \ (n = 0, 2, 4 \dots)$$

B

$$a_n = 0 \ (n = 1, 3, 5 \dots)$$

C

$$b_n = 0 \ (n = 2, 4, 6 \dots)$$

D

$$b_n = 0 \ (n = 1, 3, 5 \dots)$$

提交

Fourier Series in Trigonometric Form

Let $f(t)$ be a periodic signal with period T_1 .

Its **fundamental angular frequency** is $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

DC component: **Amplitude of cosine component :** **Amplitude of sine component:**

$$a_0 = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) dt \qquad a_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \cos n\omega_1 t dt \qquad b_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \sin n\omega_1 t dt$$

The integral limits in the above formulas are generally set as: $0 \sim T_1$ or $-\frac{T_1}{2} \sim \frac{T_1}{2}$

The Fourier series in trigonometric form can also be expressed as:

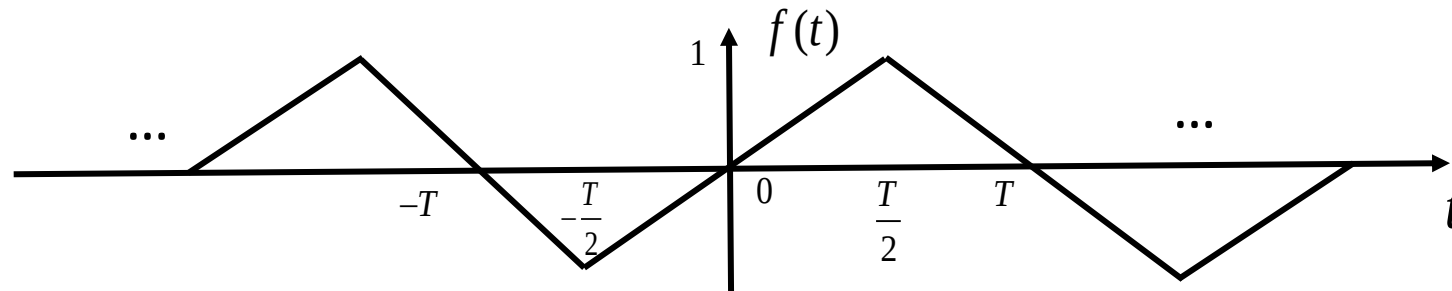
$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$

where $c_0 = a_0$ $c_n^2 = a_n^2 + b_n^2$ $\varphi_n = \arctan(-\frac{b_n}{a_n})$

Relationship Between Waveform Symmetry and Harmonic Characteristics

If $f(t)$ is a real function and its waveform satisfies a certain **symmetry**, some terms in the Fourier series disappear.

Waveform Symmetry	Frequency Components
Even Function $f(t) = f(-t)$	No sine components. $b_n = 0 \ (n = 1, 2, 3 \dots)$
Odd Function $f(t) = -f(-t)$	No DC and cosine components. $a_n = 0 \ (n = 0, 1, 2, 3 \dots)$
Odd Harmonic Function (Half-Wave Odd Symmetry) $f(t \pm \frac{T_1}{2}) = -f(t)$	No DC and even harmonic components. $a_n = 0 \ (n = 0, 2, 4, 6 \dots)$ $b_n = 0 \ (n = 2, 4, 6 \dots)$
Even Harmonic Function (Half-Wave Even Symmetry) $f(t \pm \frac{T_1}{2}) = f(t)$	No odd harmonic components. $a_n = 0 \ (n = 1, 3, 5 \dots)$ $b_n = 0 \ (n = 1, 3, 5 \dots)$



Solution: It is an odd function, so the DC and cosine components are zero.

$$a_n = 0 \quad (n = 0, 1, 2, 3 \dots)$$

It is an odd harmonic function (half-wave odd symmetry), $f(t) = -f(t \pm \frac{T}{2})$ so the DC and even harmonic components are zero.

$$a_n = 0 \quad (n = 0, 2, 4 \dots)$$

$$b_n = 0 \quad (n = 2, 4, 6 \dots)$$

Problem 3-2: Given $f(t) = u(t + 2) - u(t - 2)$, find the Fourier transform of $e^{-jt}f(t)$.

☒ A $F(\omega) = 4Sa(2(\omega + 1))$

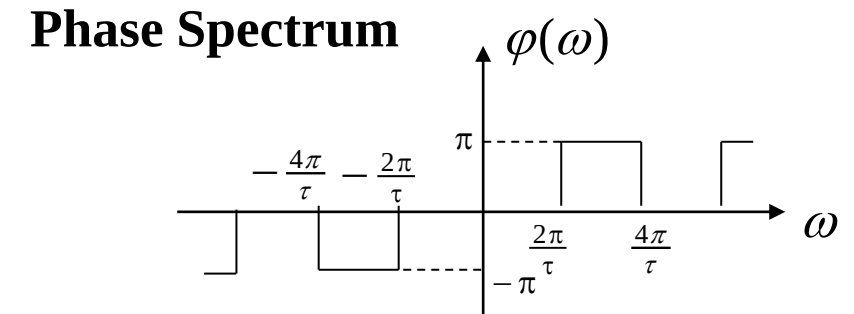
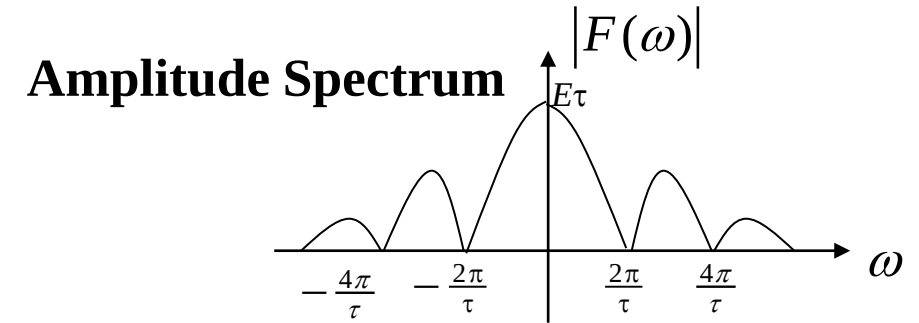
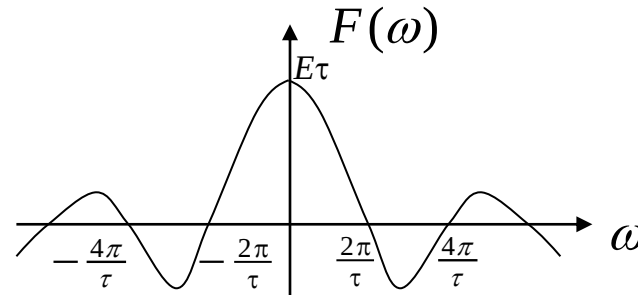
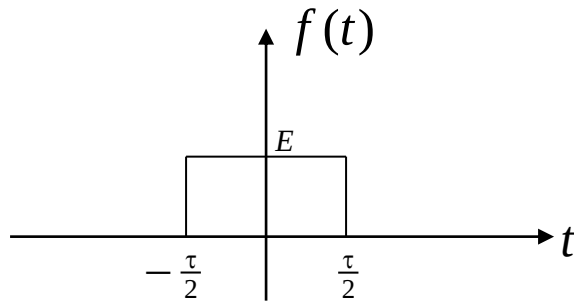
☐ B $F(\omega) = Sa(\frac{\omega + 1}{2})$

☐ C $F(\omega) = 4Sa(2(\omega - 1))$

☐ D $F(\omega) = Sa(\frac{\omega - 1}{2})$

Submit

$$E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})] \xleftrightarrow{FT} E\tau \text{Sa}(\omega \frac{\tau}{2})$$



Equivalent Bandwidth

It can be seen from the above spectrum diagram of the rectangular wave that the signal energy is mainly concentrated within the frequency range $0 \sim 2\pi / \tau$. This range, $B\omega$ is called the **equivalent bandwidth** of the rectangular wave.

$$B_{\omega} = \frac{2\pi}{\tau} \text{ (rad/s) or } \frac{1}{\tau} \text{ (Hz)}$$

Properties of Fourier Transform

(1) Time-Shift Property

If $F(\omega) = \mathcal{F}[f(t)]$ Then $\mathcal{F}[f(t - t_0)] = F(\omega)e^{-j\omega t_0}$

$$\mathcal{F}[f(t + t_0)] = F(\omega)e^{j\omega t_0}$$

(2) Frequency-Shift Property (Modulation Principle)

If $\mathcal{F}[f(t)] = F(\omega)$, then:

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

$$\mathcal{F}[f(t)e^{-j\omega_0 t}] = F(\omega + \omega_0)$$

Problem 3-2: Given $f(t) = u(t + 2) - u(t - 2)$, find the Fourier transform of $e^{-jt} f(t)$.

Solution:

Step 1: Find the Fourier Transform of $f(t) = u(t + 2) - u(t - 2)$.

$$F[f(t)] = 4\text{Sa}(2\omega)$$

Step 2: Apply the Frequency Shift Property.

$$F[e^{j\omega_0 t} f(t)] = F(\omega - \omega_0)$$

For e^{-jt} , we have $\omega_0 = -1$

$$\text{Thus } F[e^{-jt} f(t)] = 4 \cdot \text{Sa}(2(\omega + 1))$$

Problem 3-3: For a modulation system shown in Fig.1, if the spectrum of $f_1(t)$ is shown in Fig.2 and $f_2(t) = \cos(\omega_0 t)$, sketch the spectrum of $f(t)$.

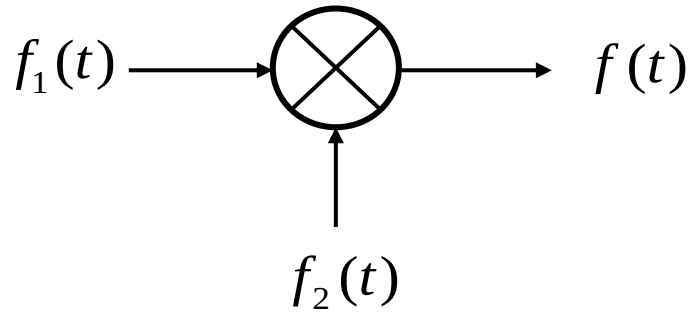


Fig.1

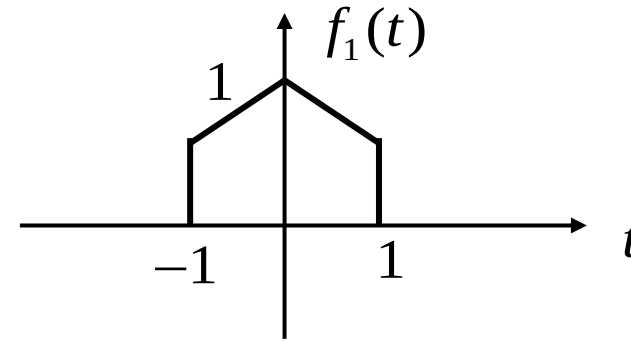


Fig.2

(1) Time-domain Convolution Theorem

The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

then

$$\mathcal{F}[f_1(t) * f_2(t)] = F_1(\omega)F_2(\omega)$$

(2) Frequency-domain Convolution Theorem

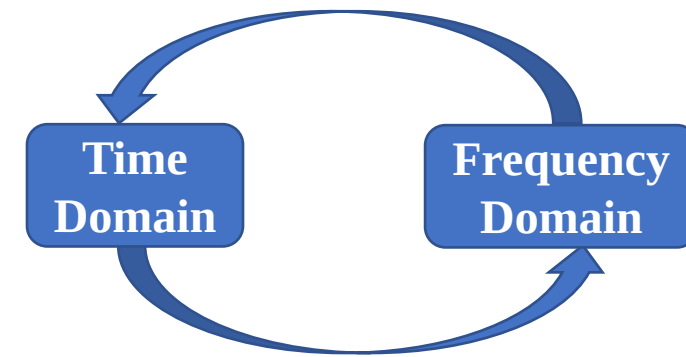
The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

then

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

$$\text{where: } F_1(\omega) * F_2(\omega) = \int_{-\infty}^{\infty} F_1(u)F_2(\omega - u)du$$



Solution:

Step 1: Find the Fourier Transform of $f_2(t) = \cos(\omega_0 t)$

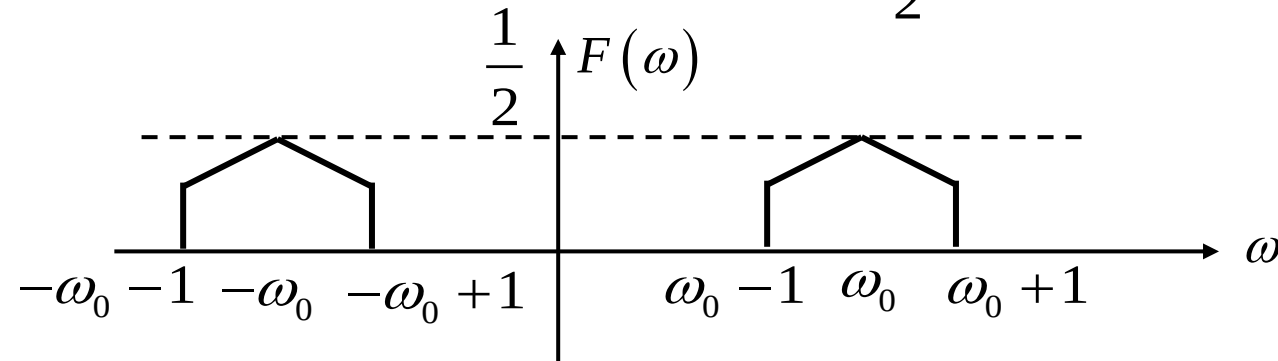
$$F_2(\omega) = F[f_2(t)] = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

Step 2: Apply the Frequency Convolution Theorem

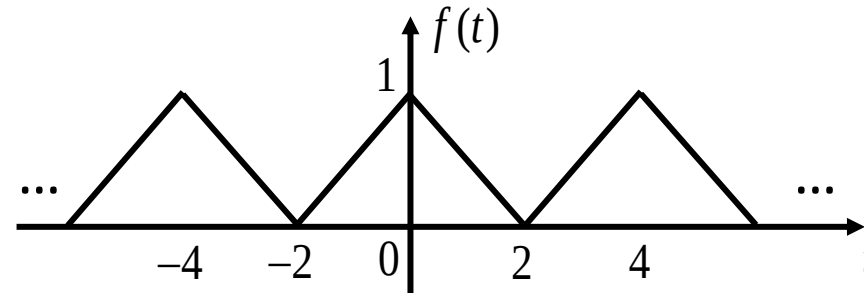
For $f(t) = f_1(t) f_2(t)$, the Fourier transform satisfies the frequency convolution theorem: $F(\omega) = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$

Substitute $F_2(\omega)$ into the formula:

$$F(\omega) = \frac{1}{2\pi} F_1(\omega) * \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] = \frac{1}{2} [F_1(\omega - \omega_0) + F_1(\omega + \omega_0)]$$



Problem 3-4: Find the Fourier Transform of the triangular periodic signal shown below.



Fourier Transform of General Periodic Signals

Let the period of the periodic signal $f(t)$ be T_1 , and the angular frequency be $\omega_1 = 2\pi f_1 = 2\pi/T_1$. Its **Fourier series in exponential form** is:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (1)$$

where
$$F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$$

Method 1: Take the Fourier transform on both sides of equation (1):

$$F(\omega) = \mathcal{F}[f(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}\right] = \sum_{n=-\infty}^{\infty} F_n \mathcal{F}\left[e^{jn\omega_1 t}\right] = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$$

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$$

Using the Frequency-shift Property

The Fourier transform of the periodic signal $f(t)$ is composed of a series of impulse functions, which are located at the harmonic frequencies of the signal ($0, \pm\omega_1, \pm2\omega_1, \dots$). The strength of each impulse is equal to 2π times the corresponding coefficient F_n of the Fourier series of $f(t)$.

$$f(t) = \sum_{-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (1)$$

Method 2: Using $F_1(\omega)$, the Fourier Transform of a single pulse signal

$$F_1(\omega) = \lim_{T_1 \rightarrow \infty} T_1 F_n$$

$$F_n = \frac{1}{T_1} F_1(\omega) \Big|_{\omega=n\omega_1} = \frac{1}{T_1} \left[\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-j\omega t} dt \right]_{\omega=n\omega_1}$$

Take the Fourier transform on both sides of equation (1):

$$F(\omega) = \mathcal{F}[f(t)] = \sum_{n=-\infty}^{\infty} F_n \mathcal{F}[e^{jn\omega_1 t}] = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$$

$$= \omega_1 \sum_{n=-\infty}^{\infty} F_1(\omega) \Big|_{\omega=n\omega_1} \delta(\omega - n\omega_1)$$

F_n is equal to the value of $F_1(\omega)$ at $n\omega_1$ multiplied by $1/T_1$ or $\omega_1/2\pi$.

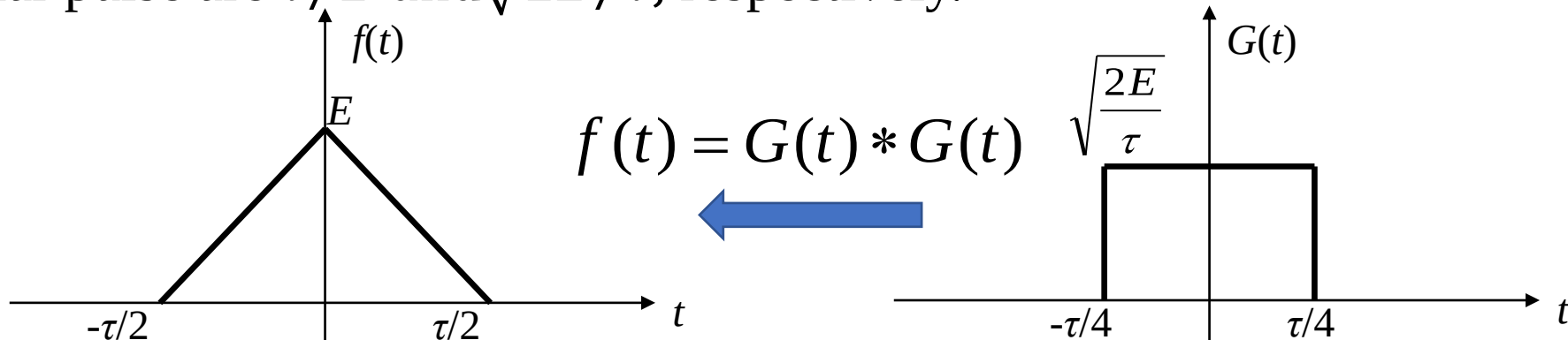
Example 3-7: Use the Time-domain Convolution Theorem to find the spectrum of a triangular pulse:

$$f(t) = \begin{cases} E(1 - \frac{2|t|}{\tau}) & |t| \leq \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$

Solution:

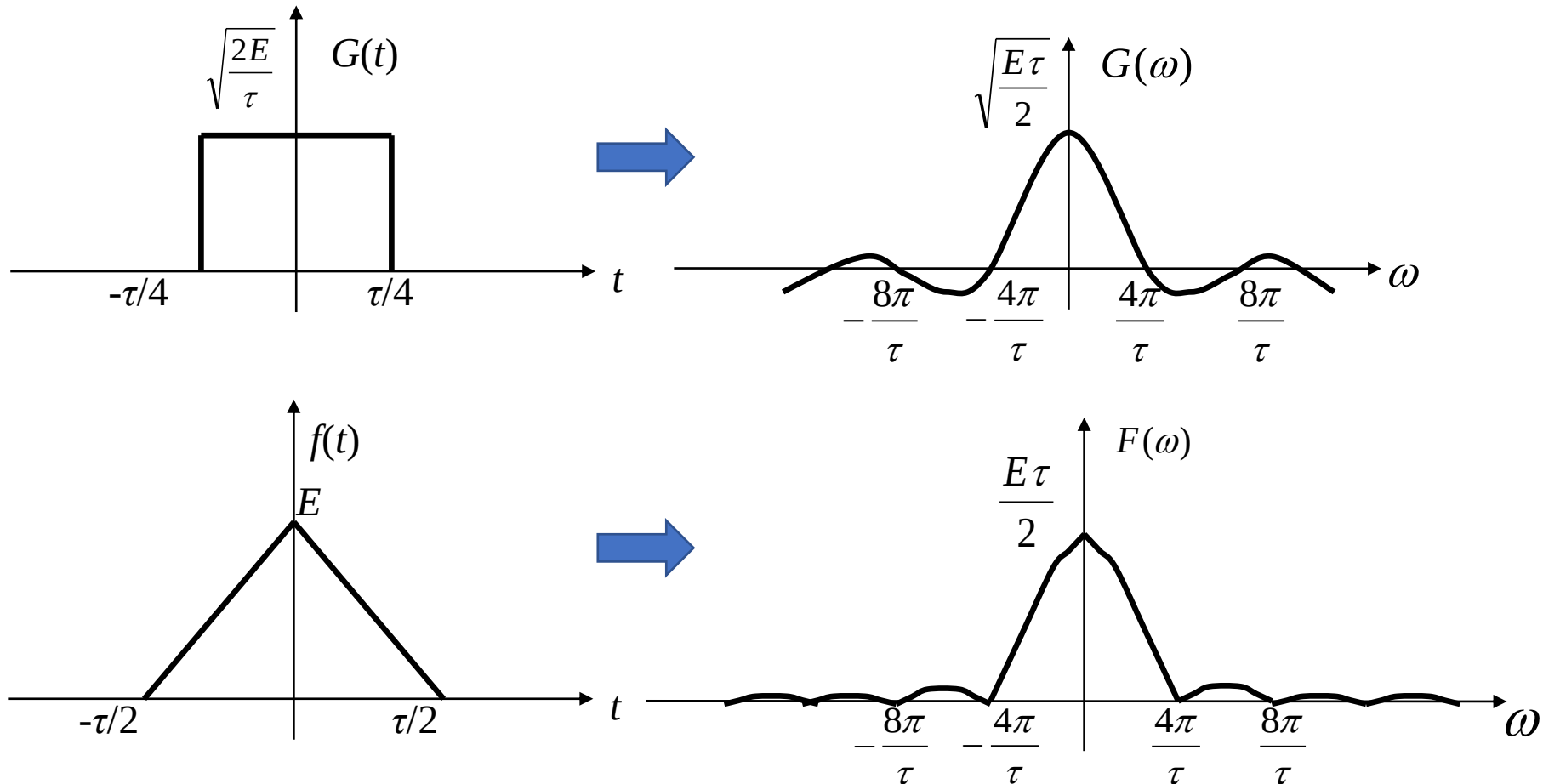
Step 1: Treat the triangular pulse as **the convolution of two identical rectangular pulses**.

The width of the triangular pulse is the sum of the widths of two rectangular pulses. The peak amplitude of the triangular pulse is the product of the square of the amplitudes of the two rectangular pulses and their width. Therefore, the width and amplitude of each rectangular pulse are $\tau/2$ and $\sqrt{2E/\tau}$, respectively.

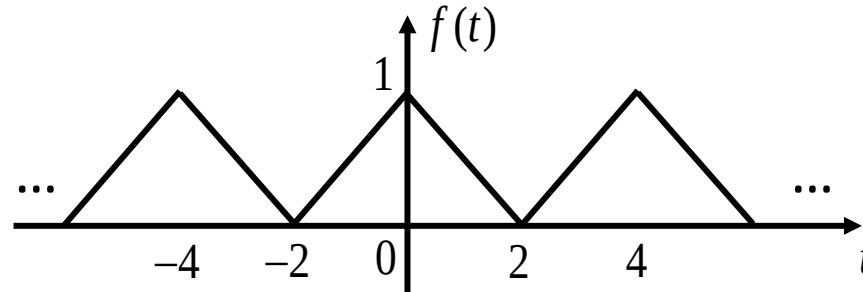


Step 2: Apply **the Time-domain Convolution Theorem**.

$$G(\omega) = \sqrt{\frac{2E}{\tau}} \cdot \frac{\tau}{2} \text{Sa}\left(\frac{\omega\tau}{4}\right) = \sqrt{\frac{E\tau}{2}} \text{Sa}\left(\frac{\omega\tau}{4}\right) \quad \therefore F(\omega) = G^2(\omega) = \frac{E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right)$$



Problem 3-4: Find the Fourier Transform of the triangular periodic signal shown below.



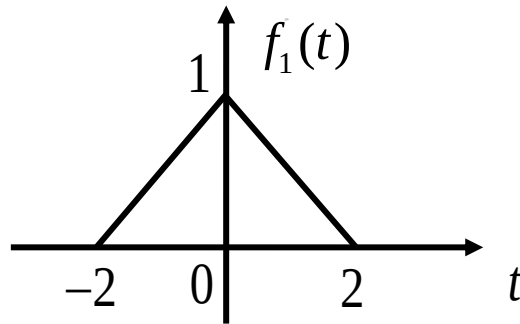
Solution:

Step 1: Determine the Period and Fundamental Frequency

The function $f(t)$ is periodic with period $T_1 = 4$. The fundamental angular frequency is:

$$\omega_1 = \frac{2\pi}{T_1} = \frac{\pi}{2}$$

Step 2: Define $f(t)$ over one Period as $f_1(t)$ and compute the Fourier Transform of $f_1(t)$.



$$E = 1, \tau = 4$$

$$F_1(\omega) = \frac{E\tau}{2} \text{Sa}^2\left(\frac{\omega\tau}{4}\right) = 2\text{Sa}^2(\omega)$$

Step 3: Compute the complex Fourier coefficients F_n for the periodic Signal.

$$F_n = \frac{1}{T_1} F_1(n\omega_1) = \frac{1}{4} \cdot 2\text{Sa}^2\left(\frac{n\pi}{2}\right) = \frac{1}{2} \text{Sa}^2\left(\frac{n\pi}{2}\right)$$

Step 4: Construct the Fourier Transform of the periodic signal.

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0) = \pi \sum_{n=-\infty}^{\infty} \text{Sa}^2\left(\frac{n\pi}{2}\right) \delta\left(\omega - \frac{n\pi}{2}\right)$$

Problem 3-5: If the highest angular frequency of $f(t)$ is ω_m , what is the maximum allowable time interval for sampling $f(t/4)$?

A $\frac{\pi}{4\omega_m}$

B $\frac{4\pi}{\omega_m}$

C $\frac{2\pi}{\omega_m}$

D $\frac{\pi}{2\omega_m}$

提交

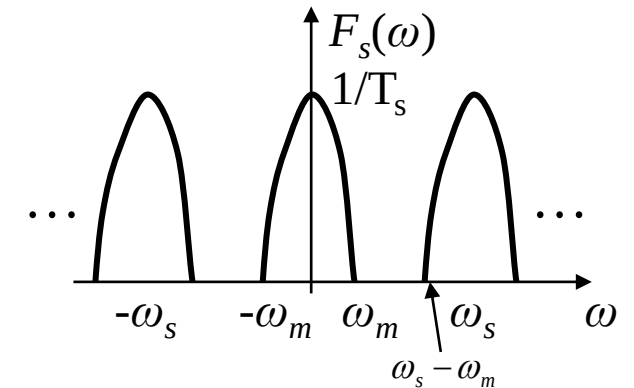
Time-domain Sampling Theorem: A band-limited signal $f(t)$, if its spectrum is confined to the range $-\omega_m \sim \omega_m$, can be uniquely represented by its equally spaced samples when the sampling interval is no larger than $\frac{1}{2f_m}$ (where $\omega_m = 2\pi f_m$) or the sampling frequency is greater than or equal to $2f_m$.

Nyquist Sampling Frequency: The minimum allowable sampling frequency.

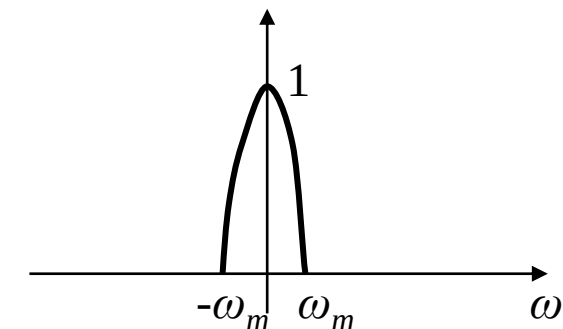
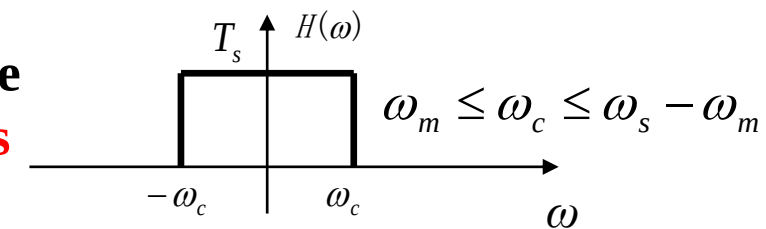
$$f_{s \min} = 2 f_m$$

Nyquist Sampling Interval: The maximum allowable sampling interval.

$$T_{s \max} = \frac{1}{f_{s \min}} = \frac{1}{2 f_m}$$



Frequency Response of the Ideal Low-pass Filter



Problem 3-5: If the highest angular frequency of $f(t)$ is ω_m , what is the maximum allowable time interval for sampling $f(t/4)$?

Solution:

According to the scaling property of Fourier transform, the signal $f(t/4)$ is expanded by 4 times in the time domain and compressed by 4 times in the frequency domain, so its highest angular frequency is $\omega_m / 4$

Then, according to the sampling theorem, the minimum sampling frequency without spectral aliasing is

$$\omega_s = 2(\omega_m / 4) = \omega_m / 2$$

The maximum sampling interval is

$$T_{\max} = \frac{2\pi}{\omega_s} = \frac{4\pi}{\omega_m}$$

Problem 4-1: Given $f(t) \leftrightarrow F(s)$, the Laplace transform of $f(2t - 6)$ is ()

A $\frac{1}{2} e^{-3s} F(2s)$

B $\frac{1}{2} e^{-3s} F\left(\frac{s}{2}\right)$

C $2e^{-3s} F\left(\frac{s}{2}\right)$

D $\frac{1}{2} e^{-s} F\left(\frac{s}{2}\right)$

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Chapter 4: Laplace Transform

Basic Properties of Laplace Transform (1)

Property	Time-domain	Complex Frequency Domain
Linearity	$\sum_{i=1}^N a_i f_i(t)$	$\sum_{i=1}^N a_i F_i(s)$
Time-Delay	$f(t - t_0)u(t - t_0)$	$e^{-st_0} F(s)$
Frequency-Shifting	$f(t)e^{-at}$	$F(s + a)$
Time-Scaling	$f(at) \quad (a > 0)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time-Domain Convolution Theorem	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Frequency-Domain Convolution Theorem	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s) / (2\pi j)$

Note: All the properties apply to **causal signals** only, i.e., the **signals $f(t)$ that satisfy $f(t) = f(t)u(t)$** .

Chapter 4: Laplace Transform

Basic Properties of Laplace Transform (2)

Property	Time-domain	Complex Frequency Domain
Time-domain Differentiation	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Time-domain Integration	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(s)}{s} + \frac{f^{-1}(0^-)}{s}$
s-domain Differentiation	$-tf(t)$	$\frac{dF(s)}{ds}$
s-domain Integration	$f(t) / t$	$\int_s^{\infty} F(\lambda) d\lambda$
Initial Value Theorem	$\lim_{t \rightarrow 0^+} f(t) = f(0^+) = \lim_{s \rightarrow \infty} sF(s)$ If $f(t)$ contains an impulse function $k\delta(t)$, $\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} s[F(s) - k]$	
Final Value Theorem	$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} sF(s)$	

Problem 4-1: Given $f(t) \leftrightarrow F(s)$, find the Laplace transform of $f(2t - 6)$.

Solution:

Step 1: Use the time scaling property.

Let $g(t) = f(2t)$,

By the **time scale property**, $g(t) \leftrightarrow \frac{1}{2} F\left(\frac{s}{2}\right)$

Step 2: Use the time shifting property.

Let $h(t) = g(t - 3) = f(2(t - 3))$

By the **time shifting property**, if $g(t) \rightarrow G(s)$

then, $g(t - t_0)u(t - t_0) \leftrightarrow e^{-st_0}G(s)$ (here $t_0 = 3$)

Thus, $h(t) \leftrightarrow e^{-3s} \cdot \frac{1}{2} F\left(\frac{s}{2}\right)$

Problem 4-2: Given the system function $H(s) = \frac{s+3}{s^2+3s+2}$, the system's excitation $x(t) = e^{-3t}u(t)$, and the initial state is zero, the system's forced response is ()

- ☒ A 0
- ☐ B $(e^{-t} - e^{-2t})u(t)$
- ☐ C $(e^{-t} + e^{-2t})u(t)$
- ☐ D $2e^{-3t}u(t)$

提交

Problem 4-2: Given the system function $H(s) = \frac{s+3}{s^2+3s+2}$, the system's excitation $x(t) = e^{-3t}u(t)$ and the initial state is zero, find the system's response, and identify the natural response and forced response.

Solution: The Laplace transform of the system's zero-state response is

$$Y(s) = X(s)H(s) = \frac{1}{s+3} \cdot \frac{s+3}{s^2+3s+2} = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

thus, the system's zero-state response is

$$y(t) = (e^{-t} - e^{-2t})u(t)$$

Poles of the system function correspond to free response components; poles of the excitation correspond to forced response components.

The response only contains the natural response component, and the forced response component is zero, due to **the pole-zero cancellation between the system function and the excitation function.**

Frequency Response from System Function Zero-pole Distribution

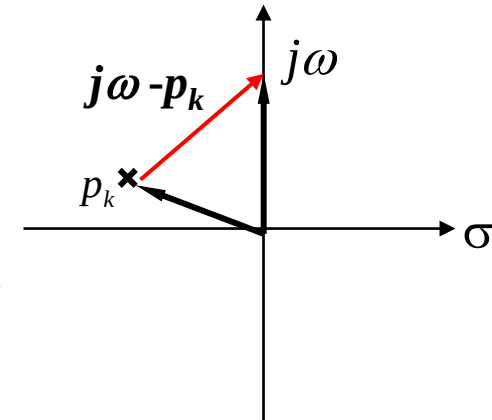
The frequency response of a system can **generally** be expressed:

$$H(\omega) = H(s) \Big|_{s=j\omega} = \frac{G \prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)} \Bigg|_{s=j\omega} = \frac{G \prod_{k=1}^M (j\omega - z_k)}{\prod_{k=1}^N (j\omega - p_k)} = \frac{G \prod_{k=1}^M B_k}{\prod_{k=1}^N A_k} e^{j(\sum_{k=1}^M \beta_k - \sum_{k=1}^N \theta_k)} = |H(\omega)| e^{j\varphi(\omega)}$$

$$|H(\omega)| = \frac{G \prod_{k=1}^M B_k}{\prod_{k=1}^N A_k}$$

$$\varphi(\omega) = \sum_{k=1}^M \beta_k - \sum_{k=1}^N \theta_k$$

G—constant



The system's magnitude-frequency response is **the ratio of the product of the magnitudes of zero vectors to that of pole vectors**.

The system's phase-frequency response is **the sum of the phase angles of zero vectors minus that of pole vectors**.

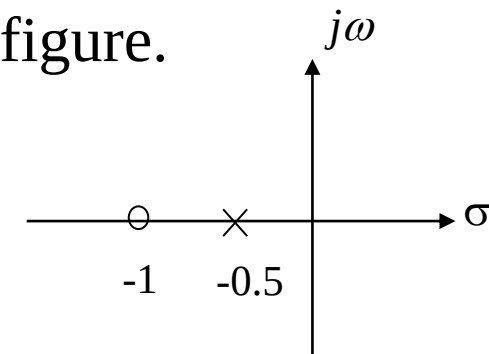
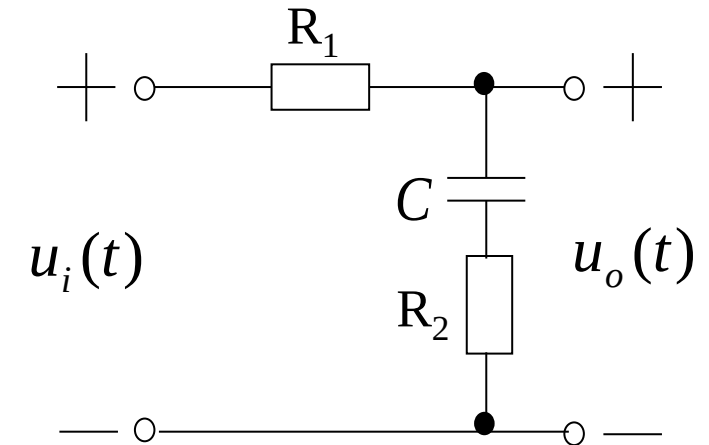
Problem 4-3: The circuit is shown in the figure, where $R_1=2\ \Omega$, $R_2=2\ \Omega$, $C=0.5\text{ F}$. Find its system function and draw its zero-pole plot. Based on the zero-pole distribution, find the frequency response, and roughly sketch the magnitude and phase frequency responses.

Solution:

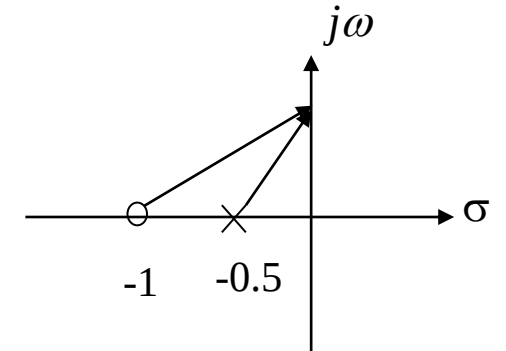
$$H(s) = \frac{U_o(s)}{U_i(s)} = \frac{\frac{1}{sC} + R_2}{R_1 + R_2 + \frac{1}{sC}} = \frac{1 + R_2 sC}{sC(R_1 + R_2) + 1}$$
$$= \frac{1 + s}{2s + 1} = \frac{1}{2} \cdot \frac{s + 1}{s + \frac{1}{2}}$$

zero location: $z_1 = -1$, Pole location: $p_1 = -\frac{1}{2}$

The zero-pole distribution on the s-plane is shown in the figure.



$$H(s) = \frac{1+s}{2s+1} = \frac{1}{2} \cdot \frac{s+1}{s+\frac{1}{2}}$$



(1) when $\omega = 0$, $|H(\omega)|=1$, $\varphi(\omega)=0$

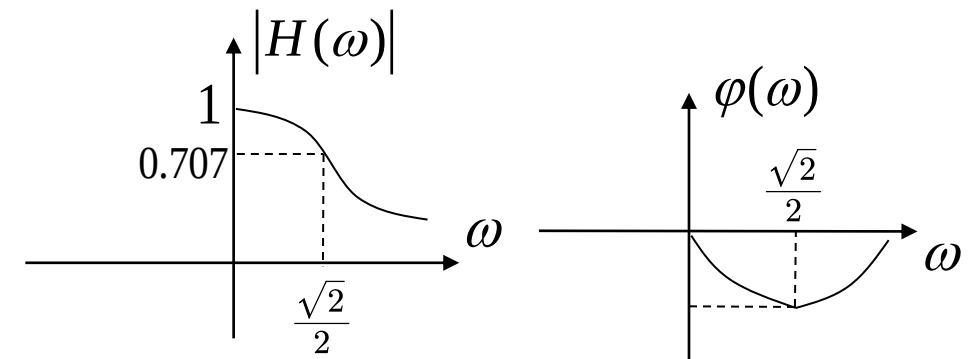
(2) when $\omega \uparrow$, the magnitude of the pole vector $\uparrow$, phase $\uparrow$; $|H(\omega)| \downarrow$, $\varphi(\omega) \downarrow$.

(3) when $\omega = \frac{\sqrt{2}}{2}$, $|H(\omega)| \approx 0.707$, $\varphi(\omega) = \arctan(\frac{\sqrt{2}}{2}) - \arctan(\sqrt{2}) \approx -19.48^\circ$

(4) when $\omega \rightarrow \infty$, $|H(\omega)| \rightarrow 1/2$, $\varphi(\omega) \rightarrow 0$.

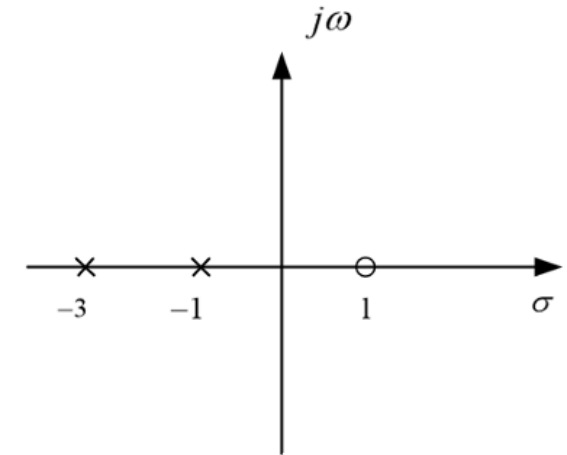
Roughly sketch the frequency response.

From its magnitude-frequency response curve, it can be seen that this is a **low-pass filter**.



Problem 4-4: Given the zero-pole distribution of the system function $H(s)$ as shown in the figure below, and $h(0_+) = 2$.

- (1) Find the expression of $H(s)$ by the initial value theorem.
- (2) Draw the simulation block diagram.
- (3) Determine the system stability.



Solution:

- (1) Zero: $z = 1$; Poles: $p_1 = -3$, $p_2 = -1$

The system function form: $H(s) = K \cdot \frac{s-1}{(s+3)(s+1)}$

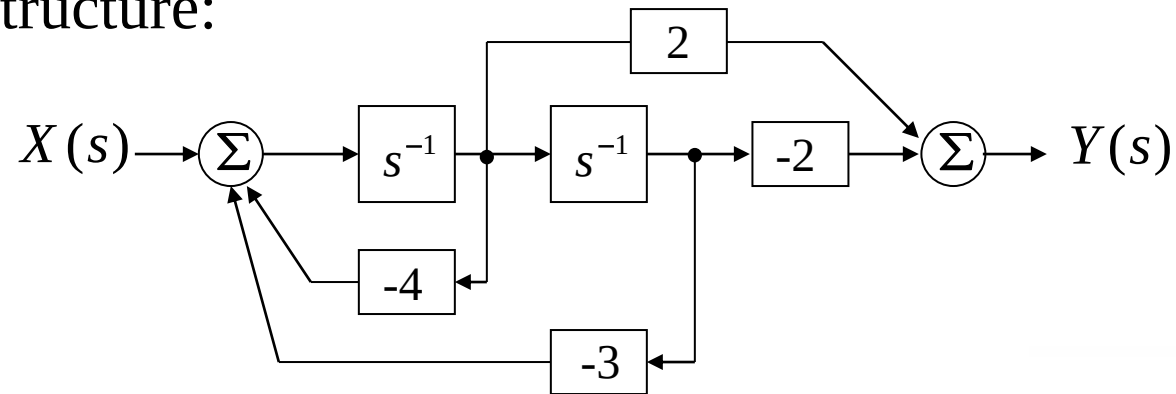
Using the **Initial Value Theorem**: $\lim_{s \rightarrow \infty} s \cdot K \cdot \frac{s-1}{(s+3)(s+1)} = K = 2$

$$\text{So, } H(s) = 2 \cdot \frac{s-1}{(s+3)(s+1)} = \frac{2s-2}{s^2+4s+3}$$

(2) From $H(s) = 2 \cdot \frac{s-1}{(s+3)(s+1)} = \frac{2s-2}{s^2+4s+3}$

Implemented with **two integrators, two summers, and four multipliers.**

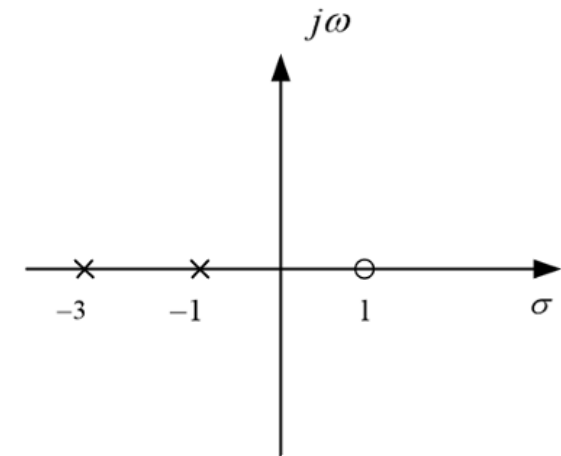
Block diagram structure:



(3) Pole locations: $p_1 = -3$, $p_2 = -1$

Both poles are on the left half plane of the s-plane.

So the system is **stable**.



Stability of Systems

Time-domain stability conditions:

The **stability** of a system is a system where **a bounded input** produces **a bounded output**. i.e., if the input $|x(\cdot)| < \infty$, the output $|y(\cdot)| < \infty$, the system must be stable. Specifically, **the output cannot contain impulse functions and their derivatives.**

The stability of an LTI system in time domain: its impulse response satisfies absolute integrability, i.e.

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

S-domain stability conditions:

- (1) All poles of the system function $H(s)$ lie on the left half-plane of the s -plane's imaginary axis; real roots are negative real numbers, and complex roots have negative real parts.
- (2) The order of the system function's numerator polynomial is no higher than that of the denominator (the number of zeros does not exceed the number of poles).

Problem 4-5: Given $F(s) = \frac{2s+3}{s(s+2)(s+3)}$, find all possible inverse bilateral Laplace transforms ().

☐ A $f(t) = \left(\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-3t} \right) u(t) \quad (\sigma > 0)$

☐ B $f(t) = -\frac{1}{2}u(-t) + \left(\frac{1}{2}e^{-2t} - e^{-3t} \right) u(t) \quad (-2 < \sigma < 0)$

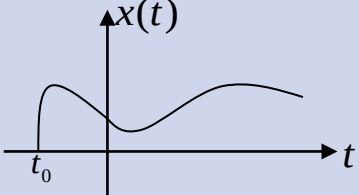
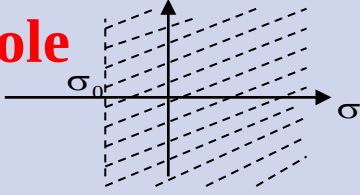
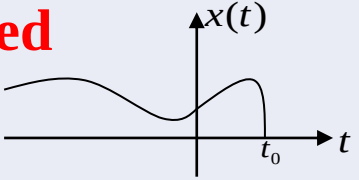
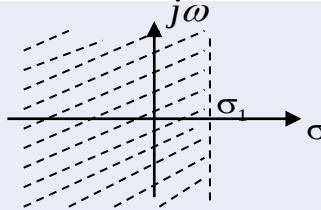
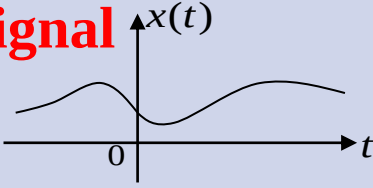
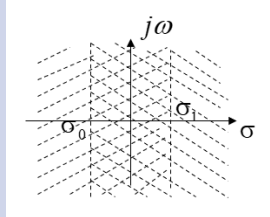
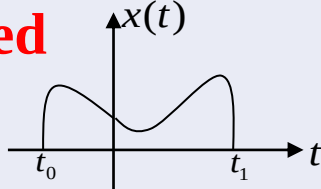
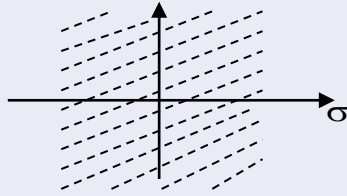
☐ C $f(t) = -\frac{1}{2}(1 + e^{-2t})u(-t) - e^{-3t}u(t) \quad (-3 < \sigma < -2)$

☐ D $f(t) = -\left(\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-3t} \right) u(-t) \quad (\sigma < -3)$

提交

Characteristics of the ROC for Bilateral Laplace Transform

When considering the existence of the ROC for the bilateral Laplace transform, **the ROC must be specified.**

Signal Category	ROC of Bilateral Laplace Transform
Right-Sided Signal 	To the right of the rightmost pole of the image function 
Left-Sided Signal 	To the left of the leftmost pole of the image function 
Bilateral Signal 	Band-shaped region between two poles (if the bilateral Laplace transform exists) 
Time-Limited Signal 	Entire s-plane 

Inverse Bilateral Laplace Transform

It can be solved based on the ROC and the poles:

- (1) Perform partial fraction decomposition on the image function;
- (2) Take the inverse transform of the image function according to the poles;
- (3) **Poles to the left of the ROC correspond to right-sided signals, and poles to the right of the ROC correspond to left-sided signals.**

F(s)	ROC	f(t)
$1/s$	$\sigma > 0$	$u(t)$
$1/s$	$\sigma < 0$	$-u(-t)$
$1/(s+\alpha)$	$\sigma > -\alpha$	$e^{-\alpha t} u(t)$
$1/(s+\alpha)$	$\sigma < -\alpha$	$-e^{-\alpha t} u(-t)$

Problem 4-5: Given $F(s) = \frac{2s+3}{s(s+2)(s+3)}$, find all possible inverse bilateral Laplace transforms.

Solution:
$$F(s) = \frac{2s+3}{s(s+2)(s+3)} = \frac{1}{2s} + \frac{1}{2(s+2)} - \frac{1}{s+3}$$

According to the poles of the image function, there are **four possible cases** for its ROC, and different ROCs will yield different time-domain forms when taking the inverse transform.

$$\sigma > 0 \quad f(t) = \left(\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-3t} \right) u(t)$$

$$-2 < \sigma < 0 \quad f(t) = -\frac{1}{2}u(-t) + \left(\frac{1}{2}e^{-2t} - e^{-3t} \right) u(t)$$

$$-3 < \sigma < -2 \quad f(t) = -\frac{1}{2}(1 + e^{-2t})u(-t) - e^{-3t}u(t)$$

$$\sigma < -3 \quad f(t) = -\left(\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-3t} \right) u(-t)$$